Common difference / ratio and explain all year

A mathematical statement has a field and assumptions. Before demonstrating it, write them; to refute it, give an example that satisfies the assumptions and contradicts the conclusion. or simulation helps to search but does not prove a universal statement.

Tool	Task and production	Accurate review
Membership and ensemble	Describe a right as a set of points; check all coordinates.	Implementation 1, p. 90.
Quantifiers and negation	Denying universal positivity and giving A witness rank.	R10; application 2.
Implications and converse	Distinguish fixed point and convergence; give a diverging sequence.	Implementation 4.
Equation	Solve a logarithmic equation After fixing the domain.	Implementation 1.
Mathematical induction	Identify property, initialize and prove Heredity to an arbitrary rank.	Applications 3 then 5.
Counter-example	Remove a hypothesis from the VIT; show this that fails.	Implementation 5.
Contrapositive	Show that two vectors of non-product None of them are orthogonal.	R60 and R61.
Absurd	Suppose four tops of a tetrahedron not flattened in a plane.	Unit 10.2.
Case breakdown	Process a null parameter before dividing.	Implementation 5.
Adequate condition	Distinguish a size guaranteed by a limit of an exact minimum size.	Applications 4 and 5.

Reusable oral protocol

In a minute: announce the object studied and the conclusion, name the tool and its assumptions, give the decisive step, then a check. The partner listens without interrupting and reformulates the conclusion ; A question then asks why a hypothesis is necessary. in each chapter are used as supports.

Short return of the listener

Is the object defined? Is the decisive hypothesis named? Is the link between the steps is the result interpreted? Choose a single point to improve before Do it again.

Results accepted and demonstrations to be reused

Operations and boundary composition, monotonous convergence, VIT, equivalences general convexity, the general existence of antiderivatives and the additivity of variance under independence are accepted in the framework indicated in progress. Evidence from the register is demonstrated and A demonstration of ITV by dichotomy, the method of Newton, oblique asymptotes and repeat combinations are deepenings possible: their knowledge is not a hidden prerequisite of the proposed evaluations.

TP supplement: compare samples

The program below extends P5·P6. The tolerance in counting is only used to absorb decimal representation of the floats; the exact calculation uses rational fractions.

```
"""Compare averages of several independent samples.""
From random    import    Random
From Statistics          import     mean, pstdev
From math    import     sqrt, comb
From fractions         import     Fraction

def   experience(n, repetitions=1000, p=0.5, seed=2026):
        if    n  <=  0 gold Repetitions < 2            or not       0  <=  p  <=  1:
              raise     ValueError(        "Out-of-domain settings"                    )
        rng = Random(grain)
        mean = [             sum(rng.random() < p            for     _ in rows       (n))/n
                             for    _ in rows       (repetitions)]
        return     {"n":  n, "average"       : average,
                    "ecart_type_observe"            : pstdev(average),
                    "ecart_type_theoretical"             : sqrt(p*(1-p)/n),
                    "proportion_ecart_at least_0_1"                  :
                          sum (abs  (m-p)  >=  0.1-1e-12 for     m in    average)/repetitions}

def   ecart_exact_equilibrium(n, delta):
        """delta is a Fraction: the entire boundaries are accurate.""
        if    n  <=  0 gold delta <= 0:
              raise     ValueError(        "n and delta must be positive"                          )
        favourable =         sum  (comb(n,k)       for     k in rows       (n+1)
                                   if abs     (Fraction(k,n)-Fraction(1.2)) >= delta)
        return     Fraction(favourable,2**n)

if   __name__==       "__main__"      :
        import    json
        print   (json.dumps([experience(n))               for     n in    (10,100,1000)],indent=2))
```

Algorithmics · P1 to P6

These programs are examples of courses. To understand them, specify entries, output and the assumptions, then execute a few steps by hand. The responses of the activities and DMs remain to be drafted separately.

P1 · Terms, sum and threshold

We assume. n positive or null integer and positive integer safety ceiling. The first function send back u_n and $\sum_{k=0}^{n} u_k$ for $u_{k+1} = au_k + b$. The second is looking for an increasing first crossing; its safety ceiling only indicates that it was not found in the number of steps set.

```
def term_sum(n, u0, a, b):
    u, total = u0, u0
    for k in rows (n):
        u = a*u + b
        Total += u
    return u, total

def Threshold(u0, a, b, target, maximum=100000):
    u, n = u0, 0
    while u <= target and n < maximum:
        u = a*u + b
        n += 1
    if u <= target:
        return None
    return n
```

P2 · Dichotomy

The calculations below use floating numbers; mathematical guarantee assumes signs Assumptions: f continues on $[a, b]$ and $f(a)f(b) \leq 0$. To identify a unique zero, separate proof of uniqueness. The function returns a length interval at the most ε, as long as machine precision still allows to share the interval.

```
def dichotomy(f, a, b, epsilon):
    assert a < b and epsilon > 0
    f, fb = f(a), f(b)
    assert fa*fb <= 0
    if Fa=0: return a, a
    if fb = 0: return b, b
    while b-a > epsilon:
        m = (a+b)/2
        if m == a gold m == b:
            raise ArithmeticError( "precision machine reached" )
        fm = f(m)
        if fm = 0: return m, m
        if f*fm < 0:
            b, fb = m, fm
        Else :
            a, fa = m, fm
    return a, b
```

P3 · Euler

With N strictly positive and $t_0 <$ end, for $y' = F(t, y)$, $Y_{k+1} = Y_k + hF(t_k, Y_k)$. The programme returns all approximations; it does not prove their accuracy.

```
def euler(F, t0, y0, fin, N):
    assert N > 0 and end > t0
    h = (end-t0)/N
    y, values = y0, [(t0, y0)]
    for k in rows (N):
        t = t0+k*h
        y = y+h*F(t, y)
        values.append((t0+(k+1)*h, y))
    return values
```

P4 · Rectangles

For N strictly positive and f continued and increasing on $[a, b]$, the result gives a limit Without monotonicity, these are only two approximations to be considered.

```
def rectangles(f, a, b, N):
    assert N > 0 and a < b
    h = (b-a)/N
    left = h* sum(f(a+k*h) for k in rows (N))
    right = h* sum(f(a+k*h) for k in rows (1, N+1))
    return left, right
```

P5 · Binomial distribution and simulation

We assume. n positive integer or zero, $0 \leq p \leq 1$ and k Integer. The direct calculation below is suitable for small sizes used in exercises; for very large parameters, libraries specialized digitals are preferable.

```
From math import comb
From random import Random

def likelihood(n, p, k):
    if k < 0 gold k > n: return 0.0
    return comb(n, k)*p**k*(1-p)**(n-k)

def cumulation(n, p, k):
    return sum (probability(n, p, j)
                for j in rows (min(k, n)+1))

def simulate_binomial(n, p, generator):
    return sum (generator.random() < p for _ in rows (n))

rng = Random(2026)
```

P6 · Fluctuations of frequency

We assume. N Total Strictly positive and $0 \leq p \leq 1$. The seed reproduces the same example. Repeat with other seeds; the law of large numbers does not say that the observed deviation decreases at each step.

```
From random import Random

def frequency(N, p, seed=2026):
    rng = Random(grain)
    success, result = 0, []
    for n in rows (1, N+1):
        success += rng.random() < p
        result.append(success/n)
    return result
```

Directory of demonstrations

This directory indicates where to find proof of course. It does not set a universal number of " Demonstrations due": groupings vary according to the presentation.

Demonstrated result	Evidence
Compare two sequences to infinity	p. 16
Growth and non-majority	p. 16
Bernoulli and geometrical limits	p. 16
Exponential limits	p. 23
Power and exponential	p. 23
Convexity and tangents	p. 37
Limit of x ln(x) to zero	p. 52
Primitive of an area, increasing case	p. 73
Integration by parties	p. 73
Pascal by count	p. 30
Pascal by calculation	p. 30
Number of parts: double counting	p. 30
Minimum distance to plane	p. 59
Cartesian Equation of Plan	p. 89
Likelihood of success	p. 44
Binomial expectation and variance	p. 81
Chebychev in a finite universe	p. 81
derivative from logarithm (differentiability permitted)	p. 52
Two antiderivatives differ from one constant	p. 67
Alternatives to $y'=ay$	p. 68
Concentration of the mean then NGL	p. 81

End-of-year and oral workshops

O1 · Transforming a topic in question

S82 · 2 h. Choose a situation that allows for program reasoning, then write a Specific question. Examples: "How to choose a number of tests to make a failure rare?"; " How does a cooling model predict a waiting time?"; "Why? Does an average become more stable when the sample grows? » Produce a page containing the the question, the assumptions, a numerical example, an accessible demonstration and a limit of the model. Allow 20 minutes of individual research, 30 minutes of mathematical work, 30 minutes of discussion by groups, 25 minutes of writing and 15 minutes of returns.

O2 · Exhibit and answer

S84 · 2 h. Prepare a 10-minute presentation and answers to the following questions: Why this model ? what hypothesis is decisive ? what does each variable mean ? what result is shown ? that changes another value of the parameter? To train in groups of three: one speaker, one interlocutor and an observer. The observer noted a clear step, a step to be justified and a formulation to be improve. Change roles. A video is useful only if it is allowed and actually reread.

O3 · Oral simulation

S87 and S93. Prepare two questions and make a presentation followed by an interview. Use a grid simple: mathematical accuracy, structure, explanation of notations, response to objections, quality The final modalities are those of the instructions and convocations of the session.